

Asya Azkin

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SUMMARY

Researcher with experience building, evaluating and productionizing end-to-end ML systems across NLP, multimodal data, and LLM applications. Skilled in Python-based ML engineering, model evaluation, and cloud deployment with a focus on automation, measurable impact, and clear communication with cross-functional stakeholders. **Strong passion for ethical, responsible AI.**

EDUCATION & SKILLS

Georgia Institute of Technology | M.S in Computer Science | Expected May 2027

Vanderbilt University | B.S. in Computer Science, Psychology

Cloud & Infra: AWS (EC2, S3, Lambda, EKS) | GCP | Docker | Kubernetes | CI/CD

ML & AI: PyTorch | TensorFlow | scikit-learn | NumPy | Pandas | MLflow | Model Monitoring | RAG Pipelines | LLM Applications (OpenAI API, Claude API, Gemini, GitHub Copilot, Fine-tuning, Prompt Engineering)

Engineering: Python | REST APIs | Feature Engineering | Model Deployment | Agentic Workflows | MCP | Git | Troubleshooting | Code Review | Agile, Scrum

EXPERIENCE & LEADERSHIP

Graduate Researcher | Georgia Institute of Technology, School of Computer Science | May 2026 – Present

Advisor: Dr. Vijay Madisetti

- Building a per-signal ablation framework for retrieval-time defenses against poisoned RAG systems, testing whether orthogonality between three filters (embedding anomaly, behavioral consistency, cross-retriever agreement) predicts robustness to adaptive attacks on standard QA benchmarks (Natural Questions, MS-MARCO) and published poisoning suites (PoisonedRAG, AgentPoison).

AI Engineer & Researcher | Emory University Democracy Lab | December 2025 – Present

Democracy Viewer - ML pipeline and web integration

- Built an **end-to-end ML pipeline and web integration on AWS** (Batch/S3) with MySQL-backed document and metadata storage, generating and evaluating **semantic embeddings for 8,000+ documents** in Python and validating clusters against domain-expert labels.
- Automated preprocessing and experimentation workflows so new datasets could be ingested and re-embedded without manual intervention, owning ML scoping and delivery across UX, research, and data science stakeholders.

Historical Document Vector Database - retrieval and evaluation infrastructure

- Built a ChromaDB Cloud vector pipeline supporting pluggable document corpora with **LLM-based summarization**, schema inference, regex-based temporal entity extraction, and metadata-filtered retrieval; **designed for fault tolerance and idempotency** via checkpoint-based resumption, batched writes, and deterministic IDs.
- Instrumented the ingestion pipeline with SQLite call logging (per-stage tokens, p50/p95/p99 latency, cost in USD) and **LangSmith tracing**, and implemented a RAGAS evaluation suite (faithfulness, answer relevancy).

Software Engineer 1 | Cox Enterprises | April 2024 - July 2025

- Built automation workflows and analytics dashboards for internal sales and operations teams, processing 10,000+ monthly transactions and **improving delivery speed by 35% through data-driven optimization**.
- Partnered with Product, Infrastructure, and stakeholder teams across 20+ interviews to **identify automation opportunities** and translate requirements into self-service data solutions.
- Implemented monitoring and testing frameworks that increased system reliability and accelerated feature delivery for business users.

PROJECTS

Who Congress Actually Listens to on Artificial Intelligence - Built an LLM-powered document analysis pipeline for 300+ government AI policy documents. Utilized OpenAI text-embedding-3-large and clustering to surface thematic groupings. Built interactive visualizations to explore co-occurrence across the corpus. [Who Congress Actually Listens to on Artificial Intelligence](#)

Media Bias & Public Perception of Artificial Intelligence - Analyzed 10,000+ global news headlines using Python, TextBlob, and scikit-learn to model sentiment and topics in AI coverage. Identified bias patterns across outlet ideology and editorial tone, examining how media framing and machine interpretation impact public trust and misinformation risk.